

A Reproducible Benchmarking Framework for Quantum Algorithms

QEncode Benchmark Suite

A Reproducible Framework for Quantum Algorithm Evaluation

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Abstract

Quantum algorithm evaluation currently lacks standardized benchmarking frameworks that allow reproducible comparison across encodings, ansätze, execution environments, and noise conditions. Experimental studies often use inconsistent configurations, making results difficult to reproduce or compare.

This whitepaper introduces the **QEncode Benchmark Suite**, a reproducible benchmarking framework for evaluating quantum algorithms with a focus on quantum chemistry workloads. The platform defines standardized benchmark specifications, automated execution pipelines, trust-certified results, workflow evaluation, and interactive analysis.

The system integrates benchmark definition, execution, certification, comparison, and visualization into a unified framework. By providing a curated benchmark suite and reproducible methodology, QEncode aims to support systematic evaluation of quantum algorithms and contribute toward the development of community benchmark standards.

1. Introduction

Benchmarking plays a critical role in evaluating the performance of computing systems. In classical computing and machine learning, benchmark suites enable fair comparison across hardware platforms, algorithms, and software systems.

In quantum computing, benchmarking efforts have primarily focused on **hardware performance metrics**, such as qubit counts, error rates, or device-level capabilities. However, standardized benchmarks for evaluating **quantum algorithms** remain limited.

Current algorithm evaluation studies often suffer from:

- inconsistent experimental configurations
- incomplete reporting of algorithm parameters
- varying measurement strategies
- differing shot budgets and noise assumptions

These differences make it difficult to compare results across experiments or reproduce findings.

To address these challenges, we introduce the **QEncode Benchmark Suite**, a framework designed to enable standardized, reproducible evaluation of quantum algorithms.

The framework integrates:

- benchmark suite definitions
- automated execution pipelines
- benchmark result databases
- trust certification mechanisms
- workflow-level algorithm evaluation
- interactive analysis tools

Together, these components provide a comprehensive benchmarking environment for quantum algorithm evaluation.

2. Design Goals

The QEncode system was designed with several key objectives.

Reproducibility

All benchmark runs must be reproducible. This is achieved through:

- deterministic benchmark configurations
- configuration hashing and validation
- standardized dataset definitions
- structured result storage

Benchmark configurations uniquely identify experiment parameters, ensuring identical experiments produce reproducible outputs.

Standardization

The framework defines a **Standard Benchmark Suite v1**, which specifies:

- benchmark problems
- algorithm configurations
- execution modes
- measurement settings
- evaluation metrics

This allows consistent comparison across experiments.

Transparency

Each benchmark result includes detailed metadata describing:

- algorithm configuration
- execution parameters
- measurement strategy

- circuit metrics
- energy estimates

This ensures full transparency of benchmark results.

Trust Certification

Benchmark results are assigned trust levels to indicate their validation status.

Level	Description
Experimental	Development or exploratory runs
Validated	Schema-compliant results with complete metrics
Certified	Results matching the official benchmark specification

Certified results represent the official benchmark outputs.

Workflow Evaluation

Rather than evaluating isolated parameters, QEncode supports **workflow benchmarking**.

A workflow represents a full algorithm strategy including:

- encoding
- ansatz
- optimizer
- measurement strategy
- mitigation method
- execution environment

Workflow benchmarking enables evaluation of practical algorithm configurations.

3. System Architecture

The QEncode platform integrates several components into a unified benchmarking framework.

Benchmark Suite Definition



Execution Pipeline



Benchmark Result Database



Trust Certification



Comparison & Ranking Engine



Visualization Dashboard

This layered architecture ensures that benchmark experiments are reproducible, traceable, and easy to analyze.

4. Standard Benchmark Suite v1

The QEncode Standard Benchmark Suite v1 defines a curated set of quantum chemistry workloads.

Benchmark Problems

Molecule	Basis	Active Space
H ₂	STO-3G	(2,2)
BeH ₂	STO-3G	(4,4)

These molecules are widely used in quantum chemistry benchmarking and remain tractable for simulation.

Encodings

The benchmark suite evaluates three fermion-to-qubit encodings:

- Jordan–Wigner
- Bravyi–Kitaev
- Parity

Each encoding provides different trade-offs in circuit complexity and qubit mapping.

Ansatz Families

Two ansatz families are included:

- UCCSD
- Hardware-Efficient Ansatz

These represent chemistry-inspired and hardware-friendly circuit constructions.

Execution Modes

Benchmarks are evaluated under three execution environments:

- Ideal simulation
- Shot-based simulation
- Noisy simulation

This enables evaluation across realistic noise conditions.

Measurement Configuration

All benchmarks use:

- grouped Pauli measurement
- fixed shot budget of 8192

These settings standardize measurement overhead.

5. Benchmark Execution Pipeline

The benchmark execution process consists of multiple stages.

Problem Construction

The molecular Hamiltonian is generated using the specified basis set and active space.

Qubit Encoding

The fermionic Hamiltonian is mapped to qubit operators using the selected encoding.

Circuit Construction

Parameterized quantum circuits are generated based on the selected ansatz.

Variational Optimization

Classical optimizers minimize the expected energy using measurement results.

Measurement

Expectation values are estimated using grouped measurement strategies.

Metric Extraction

The system records several metrics including:

- VQE energy estimate
 - exact reference energy
 - energy gap
 - circuit depth
 - two-qubit gate count
 - measurement group count
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6. Benchmark Result Database

Benchmark results are stored in a structured database.

Each entry includes:

- configuration metadata
- execution parameters
- result metrics
- trust level
- provenance information

Configuration hashes uniquely identify benchmark runs, enabling caching and result reuse.

7. Workflow Evaluation

The platform supports reusable algorithm workflows.

Example workflows include:

- `vqe_standard`
- `vqe_fast`
- `vqe_noise_resilient`

Each workflow defines a full algorithm configuration including encoding, ansatz, optimizer, measurement strategy, and mitigation method.

Workflow evaluation allows systematic comparison of algorithm strategies.

8. Benchmark Comparison and Ranking

QEncode includes automated comparison tools that evaluate benchmark results across multiple dimensions.

Metrics considered include:

- energy accuracy
- circuit depth
- two-qubit gate counts
- noise sensitivity

Ranking functions identify:

- best accuracy configuration

- lowest circuit cost configuration
- best balanced workflow

These rankings enable systematic evaluation of algorithm trade-offs.

9. Visualization and Interactive Analysis

The platform provides a dashboard for exploring benchmark results.

Users can:

- browse certified benchmark results
- compare algorithm workflows
- analyze cost vs accuracy trade-offs
- export benchmark tables

The interface includes three primary views:

- Standard Benchmark Suite
 - Workflow Comparison
 - Data Exploration
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10. Discussion

The QEncode Benchmark Suite addresses several challenges in quantum algorithm evaluation.

By providing standardized benchmark definitions, reproducible execution pipelines, and trust-certified results, the platform enables systematic algorithm evaluation.

The system is extensible and can incorporate additional workloads including:

- larger quantum chemistry systems
- optimization benchmarks
- quantum machine learning tasks

Future work will explore expanding benchmark coverage and incorporating real hardware experiments.

11. Conclusion

This whitepaper introduces the **QEncode Benchmark Suite**, a reproducible benchmarking framework for evaluating quantum algorithms.

The system integrates benchmark definitions, automated execution pipelines, trust certification, workflow evaluation, and interactive analysis into a unified platform.

By enabling transparent and reproducible benchmarking, QEncode provides a foundation for systematic evaluation of quantum algorithms and may serve as a step toward community benchmark standards.